Aquathane R100

Polyurethane liquid membrane for waterproofing and protection

DESCRIPTION

Single component high-quality polyurethane coating that cures by reacting with the humidity in the atmosphere to form a strong elastic film with excellent adhesion to different substrates. Can be applied by brush, roller or airless spray machine.

The product is based on pure elastomeric hydrophobic polyurethane resin with special inorganic filler that provides the material with excellent weathering resistance properties.

Aquathane R100 is designed to have excellent adhesion on all common construction substrates such as dry concrete, fibrous cement, ceramic tiles, wood, and galvanized steel when used in conjunction with Aquathane Primer Range.

APPLICATIONS

Waterproofing and protection of:

- Roofs.
- Light roofing made of metal or fibrous cement.
- Wet areas, bathrooms and kitchens.
- Verandas and balconies.
- Gypsum and cement boards.
- Grey and white colours provides resistance to root penetration, making it ideal for use in planters and green roofs.
- Retaining walls and basements.

ADVANTAGES

- Excellent adhesion to all common primed substrates.
- Excellent water and UV resistance. The white colour reflects much of the solar energy reducing the internal temperature of the building.
- Excellent thermal resistance. Max service temperature 80°C, max shock temperature 200°C.
- Cold Resistance: the film remains elastic down to minus 40°C.
- Excellent mechanical properties.
- Good breathability characteristics which minimize the accumulation of humidity under the coat.
- Does not rot or support mould growth.

TECHNICAL PROPERTIES @ 25°C, 55% RH:

Colour:	Grey and white
Specific gravity:	1.4 ± 0.5
Tack free time:	4 - 8 hr
Recoatable time:	6 - 24 hr
Light pedestrian traffic time:	24 - 48 hr
Final curing time:	7 days
Service temperature:	-40 to 80°C
Tensile strength: ASTM D412	≥ 5 MPa
Bond strength: ASTM D4541	≥ 1 MPa
Elongation: ASTM D412	≥ 500%
Water vapour transmission: ASTM E96	0.8 g/m².day
QUV accelerated weathering: ASTM G53	Pass (2000 hr)
Hydrolysis (8% KOH, 15 days @ 50°C):	Pass
Water permeability: (5 bars) DIN 1048	0 mm penetration
Resistance to roots: PD CEN/TS 14416:2014	No penetration

LIMITATIONS

- Only white and light grey colours can be used for exposed areas.
- Do not use on unsound substrates.
- Not recommended for waterproofing of swimming pools surfaces in contact with chemically treated water.
- Since Aquathane R100 cures with moisture, low humidity conditions will extend the tack free time and recoat time.

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Aquathane R100

STANDARDS

Aquathane R100 complies with ASTM C836.

METHOD OF USE

SURFACE PREPARATION

The surface should be clean, dry, sound and free from oil, grease and wax contamination. Cement laitance, loose particles, mould release agent or curing membranes must be removed.

Fill surfaces irregularities with a suitable product. Maximum moisture content should not exceed 5%. New concrete structures need to dry for at least 28 days.

PRIMING

In general, Aquathane R100 can be applied directly over well-prepared and sound concrete substrates. However, for highly porous surfaces, it is recommended to prime the substrate using water-based epoxy primer Aquathane Primer W or solvent-based polyurethane primer Aquathane Primer PU.

For applications on other types of substrates, please contact DCP Technical Department for best priming option.

APPLICATION

To achieve the required consistency for all application methods, including airless spray, dilute Aquathane R100 with 5 - 10% of DCP PU Solvent (consult DCP Technical Department for further details). For any mixing done on-site, low speed (300 rpm) mixer or electric drill should be used.

Apply the material with roller or brush. Apply at least two coats. Do not leave more than 24 hours between coats.

CONSUMPTION

- First coat: 0.75 - 0.90 kg/m².
- Second coat: 0.75 - 0.90 kg/m².
- Total consumption: 1.5 - 1.8 kg/m² to give 1 mm dry film thickness.

CLEANING

Clean all tools after finishing with paper towels and then wipe by using DCP solvent PU. Do not try to clean rollers.

PACKAGING

Aquathane R100 is available in 25 kg packs.

STORAGE

Aquathane R100 has a shelf life of 12 months from date of manufacture if stored in the original unopened pails at temperatures between 5°C and 25°C.

If these conditions are exceeded, contact DCP Technical Department for advice.

CAUTIONS

HEALTH AND SAFETY

Apply in well ventilated areas. Do not smoke. Do not apply near naked flames. In closed areas use force ventilation and carbon active masks. Keep in mind that solvents are heavier than air so vapour concentratiiion is higher in air closer to floor.

For further information, refer to the Material Safety Data Sheet.

FIRE

Aquathane R100 contains volatile flammable solvents.

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Note:
We endeavour to ensure that any information, advice or recommendation we may give in product literature is accurate and correct. However, because we have no control over where and how products are applied, we cannot accept any liability arising from the use of the products.

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